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Smart Safety Companion (MIT App Inventor)

Smart Safety Companion – Prototype Explanation

The **Smart Safety Companion** is a mobile application prototype developed using **MIT App Inventor**. This application is designed to help users during emergency situations by **monitoring phone movement**, **detecting strong shaking**, and **generating emergency alerts along with location and direction information**.

1. Purpose of the Prototype

The main objectives of this prototype are:

- To provide basic **personal safety support**
- To continuously **monitor phone movement**
- To detect **sudden or abnormal shaking**
- To track **GPS location**
- To identify **device direction**
- To generate and display **emergency alert messages**
- To store and use **emergency contact numbers**

This prototype works as a **safety monitoring companion** that can react automatically during risky situations.

2. User Interface Screen Description

a) Application Title

"Smart Safety Companion"

This label is displayed at the top of the screen and represents the name of the application.

b) Emergency Contact Input Section

- Text boxes are provided for entering **emergency contact numbers**
- These numbers are stored using **TinyDB**
- Stored contacts remain saved even after the app is closed
- These contacts are later used to receive emergency alerts

c) Start Monitoring Button (Green)

- Starts the safety monitoring process
- Enables:
    - Accelerometer sensor
    - Location sensor
    - Orientation sensor
- Changes the status label to **"Monitoring"**
- Begins real-time tracking of phone movement

d) Stop Monitoring Button (Red)

- Stops all monitoring activities
- Disables all sensors
- Resets shake values
- Changes the status back to **"Stopped"**
- Allows user control over the monitoring process

3. Status and Sensor Information Display

The prototype displays live data using labels:

• Status Label

Shows current system state:

- Stopped
- Monitoring
- Emergency Detected

• Accelerometer Values (X, Y, Z)

- Displays motion values from:
    - X-axis
    - Y-axis
    - Z-axis
- Used to calculate the shake intensity of the phone

• **Shake Value**

- Calculated using:
- |X| + |Y| + |Z|
- Represents overall phone movement
- High values indicate strong shaking

• **Location Information**

- Displays:
  - Latitude
  - Longitude
- Collected using GPS through LocationSensor
- Helps identify the user's current position

• **Direction Information**

- Determined using OrientationSensor
- Converts azimuth angle into readable direction:
  - North
  - East
  - South
  - West

• **Alert Message Display**

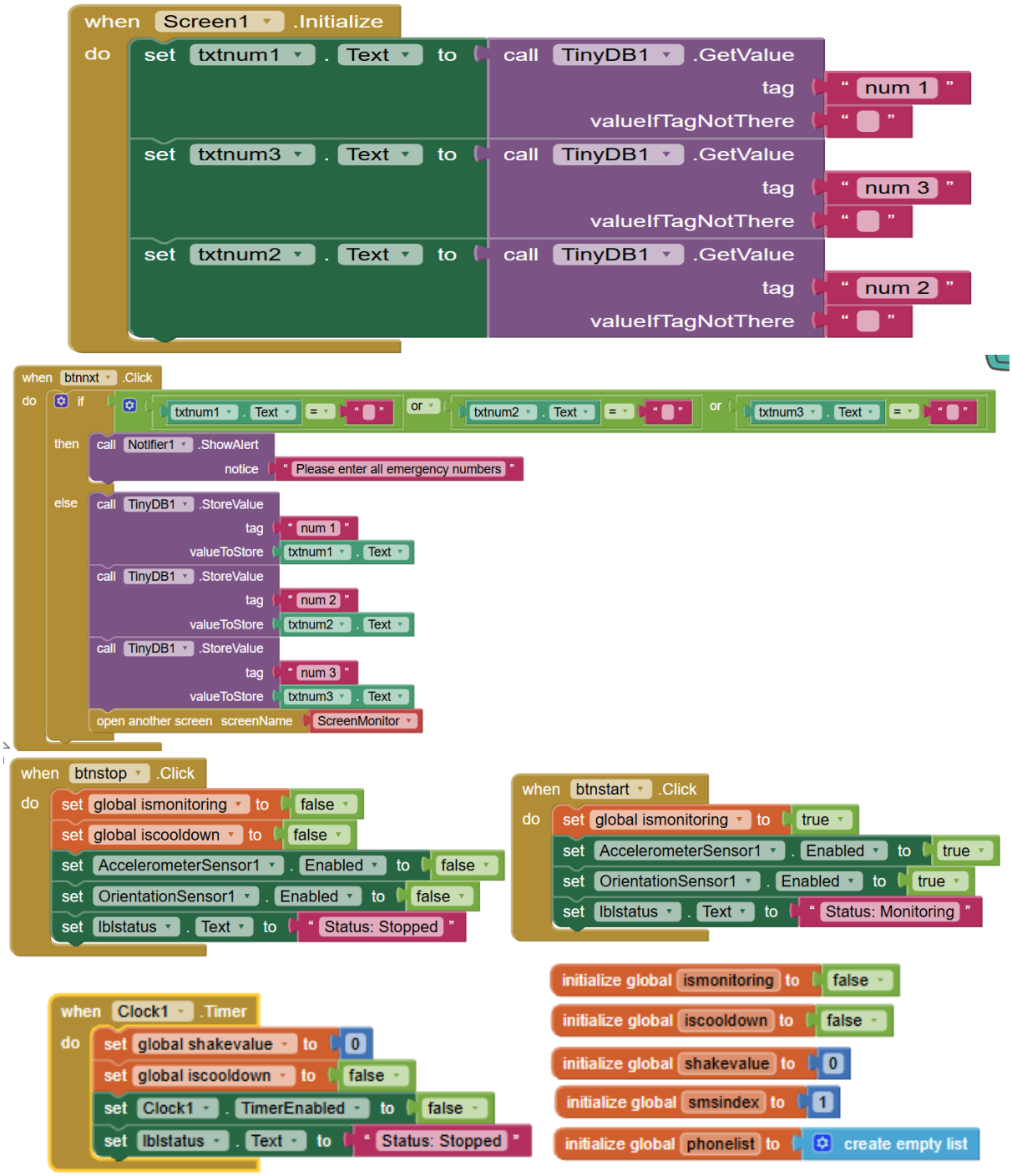
- Shows emergency alert messages
- Message includes:
  - Emergency warning
  - Location
  - Direction
  - Time
- Updates automatically when danger is detected

4. Background Design

- A technology-themed background image is used
- Improves visual appearance
- Makes the app more attractive and user-friendly

5. Block Programming Description (MIT App Inventor)

The prototype logic is created using **MIT App Inventor Blocks**.



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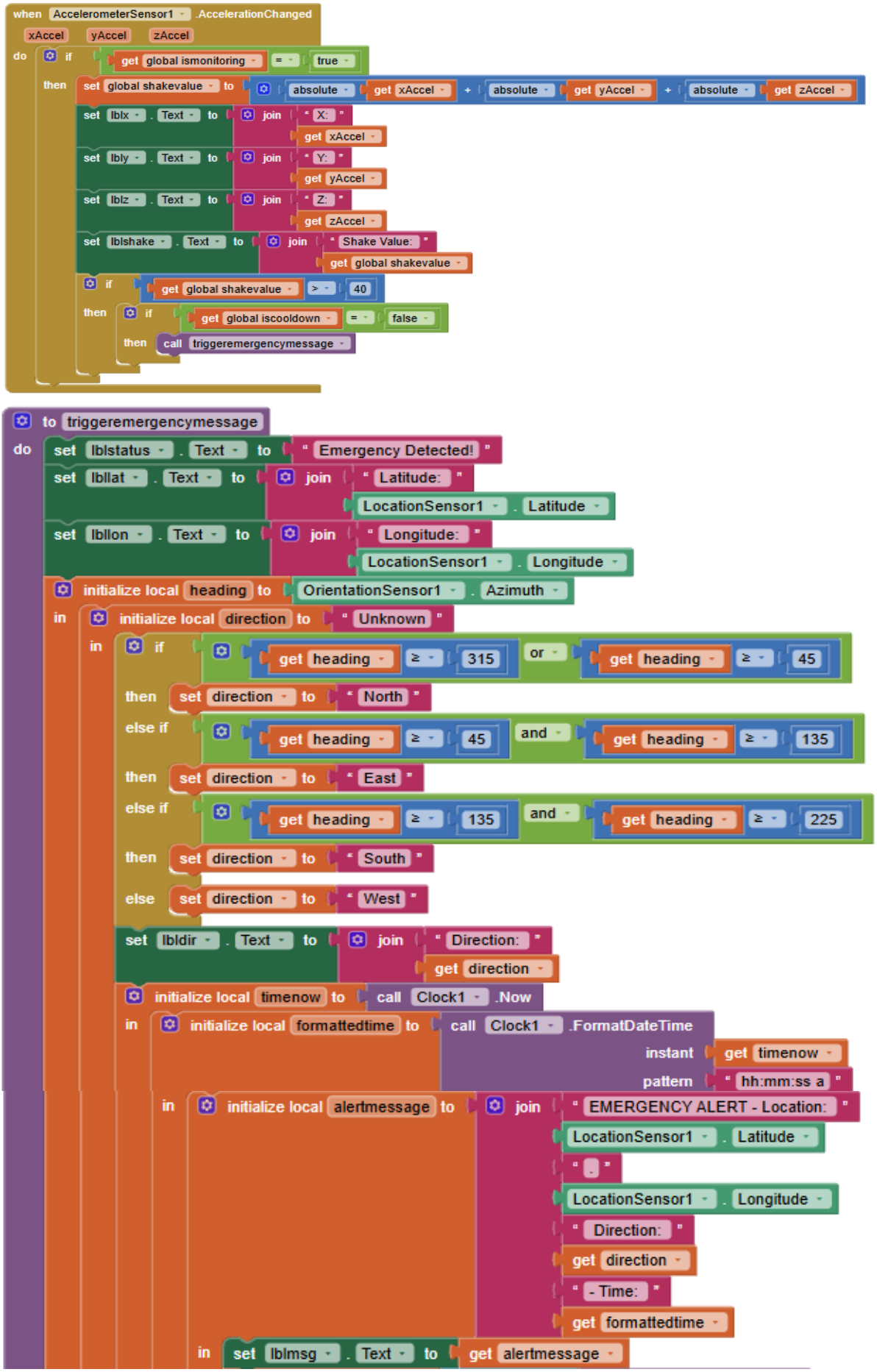
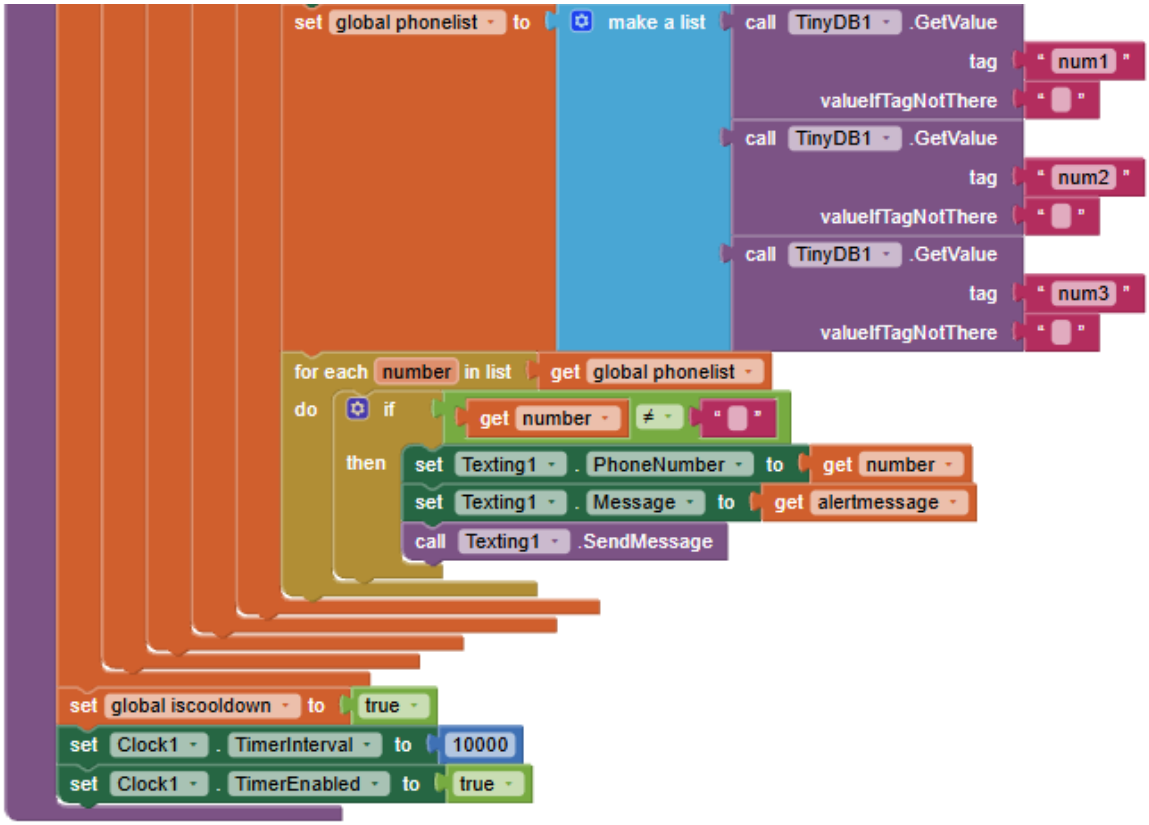




Global Variables

The app uses global variables to control its behavior. Monitoring decides whether the system is active, cooldown prevents repeated alerts, shakeThreshold sets the minimum movement needed to trigger an emergency, directionText stores the detected direction, and alertMessage holds the generated emergency message.

Accelerometer Sensor:

The accelerometer runs whenever the phone moves. It checks if monitoring is active and cooldown is off, then calculates the shake value from X, Y, and Z readings while showing live sensor data. If the shake value exceeds the threshold, an emergency is detected, the status changes to "Emergency Detected," the location and direction are determined, a message is generated, cooldown is activated, and the clock timer begins.

Direction Detection Logic

Direction is determined using the azimuth angle. Values between 315° and 45° indicate North, 45° to 135° indicate East, 135° to 225° indicate South, and 225° to 315° indicate West. This converts raw sensor readings into simple, human-readable directions.

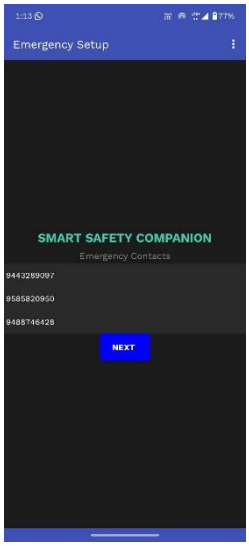
Clock Timer Block

- Timer runs for 10 seconds,Prevents repeated alerts
- After timer ends:
    - Cooldown resets

- App returns to monitoring mode

6. Overall Working of the Prototype

1. User enters emergency contact numbers
2. User presses **Start Monitoring**
3. App continuously monitors phone movement
4. If strong shaking occurs:
    - Emergency is detected
    - Location and direction are captured
    - Alert message is generated and displayed
5. Cooldown prevents multiple alerts
6. User can stop monitoring anytime



7. Conclusion

The **Smart Safety Companion prototype** demonstrates a simple yet effective **safety monitoring system** using smartphone sensors.
It shows how movement detection, location tracking, and direction awareness can be combined to assist users during emergencies.

This prototype can be extended further for:

- Women safety
- Accident detection
- Elderly assistance
- Emergency response systems.